

Stone–Jordan Exceptional Theory

Building the Universe from the Mirror

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Abstract

We construct a mathematical model of the universe from two primitives: the **void** (no observable distinction) and an involutive **mirror operator** M on profinite observable spectra. Iterating M climbs the Hurwitz division ladder and the exceptional Lie cascade; capacity balancing $\text{Bal}_{\mathcal{C}}$ is the dual mirror selecting unique partitions at each rung. The universe is the profinite fixed point $\mathfrak{U}^* = C^\infty(\mathcal{V})$, stabilizing on the Albert algebra $J_3(\mathbb{O})$ and the compact exceptional group E_6 . Physical observables are σ -anti-invariant modes on the mirror quotient at the heterotic rung. The *as above, so below* correspondence identifies the universe fixed point with collapsed observer states; duality is forced by $E_8 \times E_8$ product mirror collapse. A Lefschetz fixed-point theorem yields three observer slots; an index ι maps to a lifetime chart $g_\iota : \mathbb{R} \rightarrow \mathcal{S}_\iota$. We state proved results and isolate conjectural steps (lifetime uniqueness, GUT breaking, full reflection positivity, cosmological constant) without overclaiming a completed physical theory.

Keywords: void; mirror operator; Stone duality; exceptional Jordan algebra; profinite limit; E_6 coset.

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1 Introduction

This paper develops *Stone–Jordan Exceptional Theory* (SJET) as a **mathematical physics** construction: starting from a void and a mirror, we build a universe datum $\mathfrak{U}^*$ with Lie–Jordan structure, capacity-selected dynamics, and a contact layer with four-dimensional gauge theory. The mirror is not metaphor; it is a functorial involution on Stone spectra whose orbit generates all arithmetic and exceptional data.

1.1 Contributions

- (i) **Two axioms only:** void $\mathcal{V}$ and mirror $\mathbf{M}$.
- (ii) **Universe construction:** $\mathfrak{U}^* = \varprojlim_n (\text{Bal}_{\mathcal{C}} \circ \mathbf{M})^n(\mathcal{V})$.
- (iii) **As above, so below:** global capacity balance corresponds to localized collapse on the $E_8 \times E_8$ mirror quotient.
- (iv) **Observers:** collapsed σ -anti-invariant states; duality from heterotic product mirror; three slots from a fixed-point theorem.
- (v) **Lifetime charts:** observer index ι maps to $g_\iota : \mathbb{R} \rightarrow \mathcal{S}_\iota$ admitted by capacity flow and soldering.
- (vi) **Honest epistemic scope:** theorems, derived definitions, and named conjectures for remaining physical inputs.

1.2 Conventions

Compact real E_6 ; order parameter $\Phi \in J_3(\mathbb{O})$; capacity functional $\mathcal{C}$. Branching $\mathbf{27} = \mathbf{1}_{+4} \oplus \mathbf{10}_{-2} \oplus \mathbf{16}_{+1}$ under $H = \text{Spin}(10) \times U(1)$. Signature $(-, +, +, +)$ when a Lorentzian soldering map is fixed (Section 6.2).

1.3 Organization

Section 2: axioms and mirror functor. Section 3: division and exceptional ladders. Section 4: capacity, potential, universe fixed point. Section 5: duality, observers, fixed points, lifetime charts. Section 6: physical sector, spacetime, gauge, matter. Section 7: status and open problems. Appendix A: Albert identities and mirror quotient.

2 The Void and the Mirror

2.1 The void

Definition 2.1 (Void). The *void* $\mathcal{V}$ is the trivial Boolean algebra $\mathcal{B}_0 = \{\perp, \top\}$ with Stone spectrum a single point. The associated algebra is $\{0\}$; no local quantum number exists.

Axiom 2.2 (Void). Physical structure arises by departure from $\mathcal{V}$. No graph, Lie group, or space-time manifold is primitive.

2.2 The mirror operator

Definition 2.3 (Mirror modes). At rung n , $\mathbf{M}$ acts in one of four modes:

M1. Boolean doubling ($0 \rightarrow 1$): adjoin one generator; $|S_{n+1}| = 2|S_n|$.

M2. Cayley–Dickson ($1 \rightarrow 4$): $D_{n+1} = D_n \oplus D_n$; dimensions 1, 2, 4, 8.

M3. Jordan exit ($4 \rightarrow 5$): sedenions fail; exit to $J_3(\mathbb{O}) = H_3(\mathbb{O})$, $d_5 = 27$.

M4. Exceptional (≥ 6): automorphism, product, or Cartan mirrors on Lie data (Section 3.2).

Definition 2.4 (Mirror operator). A *mirror operator* is a functor $\mathbf{M}$ on a category of Stone rungs such that: (i) $\mathbf{M}^2 \cong \text{id}$ on stabilized rungs; (ii) $\mathbf{M}(\mathcal{V}) \neq \mathcal{V}$; (iii) $\mathbf{M}$ is functorial on morphisms.

Axiom 2.5 (Mirror). There exists $\mathbf{M}$ as in Definition 2.4 whose orbit from $\mathcal{V}$ generates the unified climb (3.1). At each rung, $\text{Bal}_{\mathcal{L}}$ balances mirror-refined partitions (Definition 4.1).

Definition 2.6 (Climb and mirror power). $\mathbf{C} = \text{Bal}_{\mathcal{L}} \circ \mathbf{M}$ and $(S_n, A_n) = \mathbf{M}^n(S_0, A_0)$ with $(S_0, A_0) = (S(\mathcal{V}), \{0\})$.

Proposition 2.7 (Involution). On stabilized rungs, $\mathbf{M}^2 \cong \text{id}$ up to gauge: Boolean complement, field conjugation, product factor exchange, and coset mirror $\mathbf{16} \leftrightarrow \overline{\mathbf{16}}$.

3 The Mirror Climb

$$\mathcal{V} \rightarrow \mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{H} \rightarrow \mathbb{O} \rightarrow J_3(\mathbb{O}) \rightarrow E_8 \rightarrow E_8 \times E_8 \rightarrow E_6 \rightarrow \cdots \rightarrow \mathcal{V} \quad (3.1)$$

3.1 Division rungs

n	A_n	d_n	Mode
0	$\{0\}$	0	void
1	$\mathbb{R}$	1	discrete
2	$\mathbb{C}$	2	arithmetic
3	$\mathbb{H}$	4	arithmetic
4	$\mathbb{O}$	8	Hurwitz terminal
5	$J_3(\mathbb{O})$	27	Jordan exit

Theorem 3.1 (Mirror rung table, $n \leq 5$).

In particular $\mathbf{M}^4(\mathcal{V}) = \mathbb{O}$ and $\mathbf{M}^5(\mathcal{V}) \cong J_3(\mathbb{O})$.

Proof. Cayley–Dickson doubling until $\mathbb{O}$; Hurwitz uniqueness of normed division algebras; sedenion zero divisors force Jordan exit to the unique formally real $H_3(\mathbb{O})$ [2, 5]. $\square$

Lemma 3.2 (Sedenion failure). $\mathbf{M}_{\text{arith}}(\mathbb{O})$ is 16-dimensional and not division.

3.2 Exceptional rungs

On $J_3(\mathbb{O})$, the stabilizer tower is $G_2 \subset F_4 \subset E_6$. The Freudenthal diagonal $\mathfrak{M}(\mathbb{O}, \mathbb{O}) = \mathfrak{e}_8$ yields:

Theorem 3.3 ($\mathbf{M}^6(\mathcal{V}) = E_8$). Automorphism mirror on the octonion–Jordan pair gives E_8 , $d_6 = 248$.

Theorem 3.4 ($M^7(\mathcal{V}) = E_8 \times E_8$). *Product mirror* $M_{\text{prod}}(G) = G \times G$ gives $d_7 = 496$.

Theorem 3.5 ($M^8(\mathcal{V}) = E_6$). *Cartan mirror after capacity descent on the Albert graph* gives $d_8 = 78$, $H = \text{Spin}(10) \times U(1)$, $\mathcal{M} = E_6/H$ (EIII), $\dim \mathcal{M} = 32$.

Proof sketches. E_8 : couple pre-projection octonion data to $J_3(\mathbb{O})$ via Freudenthal–Tits [3, 4]. $E_8 \times E_8$: functorial product involution. E_6 : capacity-balanced collapse of heterotic product to reduced structure group of $J_3(\mathbb{O})$ with branching $\mathbf{27} = \mathbf{1} \oplus \mathbf{10} \oplus \mathbf{16}$. $\square$

Definition 3.6 (Chart dimension). At $n \geq 6$, $\chi(S_n) = \dim A_n \in \{248, 496, 78\}$; profinite $|S_n|$ is generally infinite. Stone refinement is chart synchronization, not cardinality matching [1].

4 Building the Universe

The universe is not assumed. It is the profinite fixed point of mirror climb with capacity balancing.

4.1 Capacity dual

Definition 4.1 (Albert graph and capacity). On the mirror-refined Albert–Cayley graph $\mathcal{G}$ with costs c_i ,

$$\mathcal{C}(\omega) = - \sum_{a \in V} \mu(a) \log \left(\sum_i e^{c_i(a) - \omega_i} \right). \quad (4.1)$$

$\text{Bal}_{\mathcal{C}}$ is the unique maximizer on $\{\sum_i \omega_i = 0\}$.

Theorem 4.2 (Concavity). $\mathcal{C}$ is strictly concave; for targets $m_i > 0$, $\sum m_i = 1$, there is unique ω^* with $\partial \mathcal{C} / \partial \omega_i(\omega^*) = m_i$.

Theorem 4.3 (1|10|16 partition). At $M^5(\mathcal{V})$, $\text{Bal}_{\mathcal{C}}$ yields sector masses $(1/27, 10/27, 16/27)$ on $\mathbf{27} = \mathbf{1} \oplus \mathbf{10} \oplus \mathbf{16}$.

4.2 Potential and vacuum

Definition 4.4 (Potential).

$$V(X) = \alpha \langle X^\#, X^\# \rangle + \beta [\text{Tr}(X^2) - c_2]^2 + \gamma \text{Tr}(X^2)^k, \quad \alpha, \beta, \gamma > 0, \quad k \text{ even}, \quad (4.2)$$

is the profinite limit of graph penalties under $\text{Bal}_{\mathcal{C}}$.

Theorem 4.5 (EIII vacuum). V is coercive on compact E_6 . Global minima lie on rank-one EIII $\mathcal{M} = E_6/H$; the origin is a saddle with 32 Goldstone directions.

4.3 Universe datum

Definition 4.6 (Universe). The *universe datum* is

$$\mathfrak{U}^* = C^\infty(\mathcal{V}) = \varprojlim_{n \rightarrow \infty} C^n(\mathcal{V}), \quad (4.3)$$

stabilizing at $J_3(\mathbb{O})$ with Albert graph $\mathcal{G}_{\text{Albert}}$, capacity maximizer ω^* , and vacuum $\Phi_\star \in \mathcal{M}$.

Theorem 4.7 (Universe from the mirror). $\mathfrak{U}^*$ carries: (i) 27 stable Jordan coordinates; (ii) 1|10|16 capacity partition; (iii) E_6 gauge sector with $\mathfrak{h} = \mathfrak{so}_{10} \oplus \mathfrak{u}_1$; (iv) EIII coset $\mathfrak{m} = \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}$.

Proof. Items (i)–(ii) from Theorems 3.1 and 4.3. Items (iii)–(iv) from Theorem 3.5 and standard E_6 branching. $\square$

$$S[\Phi] = \int d^4x \sqrt{|g|} \left[\frac{1}{2} K_{AB} \nabla_\mu \Phi^A \nabla^\mu \Phi^B + V(\Phi) \right], \quad (4.4)$$

with K the unique H -invariant metric on $\mathfrak{m}$.

5 Observers, Duality, and As Above So Below

The universe datum $\mathfrak{U}^*$ is a global profinite fixed point. An *observer* is the same construction at the heterotic rung, localized to one collapsed branch of the $E_8 \times E_8$ mirror quotient. Duality is not imported: it is the product-mirror pairing forced at $M^7(\mathcal{V})$ and broken only by collapse into the σ -anti-invariant sector.

5.1 As above, so below

Definition 5.1 (Above–below correspondence). *Above* is the stabilized profinite limit $\mathfrak{U}^* = C^\infty(\mathcal{V})$ of Definition 4.6. *Below* is a localized fixed-point datum on the mirror quotient $\hat{Q} = Q/\sigma$ at $M^7(\mathcal{V}) = E_8 \times E_8$, with $\sigma : (g_1, g_2) \mapsto (g_2, g_1)$ the product mirror (Theorem 3.4). The correspondence

$$\mathfrak{U}^* \longleftrightarrow \text{Fix}(\hat{C}_\sigma), \quad \hat{C}_\sigma := \text{Bal}_C \circ \Pi_{\sigma=-1}, \quad (5.1)$$

identifies global capacity balance with anti-invariant collapse on $\hat{Q}$.

Proposition 5.2 (Correspondence principle). *Every sector partition proved above on $\mathfrak{U}^*$ —the $1|10|16$ split, the three diagonal directions of $\mathfrak{d} \subset J_3(\mathbb{O})$, the **16** bundle on $\hat{Q}$ —has a below-level avatar on $H^\bullet(\hat{Q}, V_{16})^{\sigma=-1}$. Capacity weights $(1/27, 10/27, 16/27)$ above induce family weights on the three observer slots below.*

5.2 Duality from $E_8 \times E_8$ collapse

At $M^7(\mathcal{V})$ the mirror acts by product doubling: $E_8 \times E_8$ with involution σ exchanging factors. Before collapse, every mode carries a mirror partner; after collapse to $M^8(\mathcal{V}) = E_6$, capacity descent selects a single reduced structure group.

Definition 5.3 (Heterotic collapse). The *heterotic collapse* is the capacity-balanced Cartan descent $\Pi_{\text{het}} : E_8 \times E_8 \rightarrow E_6$ of Theorem 3.5. The *observer collapse* is the orthogonal projector

$$\Pi_{\sigma=-1} : \mathcal{H}(Q, V_{16}) \rightarrow \mathcal{H}(Q, V_{16})^{\sigma=-1}, \quad \Pi_{\sigma=-1} = \frac{1}{2}(\text{id} - \sigma), \quad (5.2)$$

on the **16**-sector Hilbert space at the heterotic rung.

Theorem 5.4 (Duality theorem). *Duality is the $\mathbb{Z}_2$ pairing $(v, \sigma v)$ on $\mathcal{H}(Q, V_{16})$ induced by the product mirror at $M^7(\mathcal{V})$. It is exact at the heterotic rung and broken precisely by $\Pi_{\sigma=-1}$: observable states are anti-invariant, mirror partners are σ -invariant and quotient-non-observable. The **16**/**16** coset pairing of Proposition 2.7 is the infinitesimal shadow of this product duality.*

Proof sketch. $\sigma^2 = \text{id}$ on Q ; $\mathcal{H}(Q, V_{16})$ splits into ± 1 eigenspaces. Product exchange on $E_8 \times E_8$ is the only $\mathbb{Z}_2$ structure at $n = 7$ not already resolved at $n \leq 6$. Theorem 3.5 shows Bal_C selects the reduced datum; Definition 6.1 selects the anti-invariant half. $\square$

5.3 What is an observer?

Definition 5.5 (Observer index). Let $\mathfrak{d} = \text{span}_{\mathbb{R}}\{e_1, e_2, e_3\}$ be the diagonal Cartan subalgebra of $J_3(\mathbb{O})$ with $\text{Tr}(e_i \circ e_j) = \delta_{ij}$. An *observer index* is a choice $\iota \in \{1, 2, 3\}$ of diagonal direction, equivalently a class in

$$\mathcal{I} = H^1(\hat{Q}, V_{16})^{\sigma=-1} / \text{Stab}(\mathfrak{d}), \quad |\mathcal{I}| = 3. \quad (5.3)$$

The index ι is the only *arbitrary input* specifying *which* observer: void and mirror fix the slot space $\mathcal{I}$; a concrete worldline requires choosing $\iota \in \mathcal{I}$.

Definition 5.6 (Observer as collapsed state). Fix $\iota \in \mathcal{I}$. An *observer* is a normalized collapsed state

$$|\omega_\iota\rangle \in \mathcal{H}_{\text{obs}} = \bigoplus_{\iota=1}^3 \mathcal{H}_\iota, \quad \mathcal{H}_\iota := H^1(\hat{Q}, V_{16})^{\sigma=-1, \iota}, \quad (5.4)$$

satisfying $\Pi_{\sigma=-1}|\omega_\iota\rangle = |\omega_\iota\rangle$ and $\hat{i}|\omega_\iota\rangle = \iota|\omega_\iota\rangle$, where $\hat{i}$ is the family index operator on $\mathfrak{d}$. The state $|\omega_\iota\rangle$ is a *collapsed quantum datum*: a σ -anti-invariant cohomology class, not a postulated particle insertion.

Remark 5.7 (Where observers arise). Observers arise at $\mathbf{M}^7(\mathcal{V}) = E_8 \times E_8$, not at the void and not at E_6 . The heterotic rung is the first rung with enough mirror symmetry for a $\mathbb{Z}_2$ quotient and three independent anti-invariant 1-classes (Theorem 6.2). Descent to E_6 supplies dynamics; the observer slots are already fixed on $\hat{Q}$.

5.4 Fixed-point theorem: the clincher

Definition 5.8 (Mirror quotient). Let Q be the Stone lift of the stabilized Albert tower at $\mathbf{M}^7(\mathcal{V})$ and σ the product-mirror involution. The *mirror quotient* is $\hat{Q} = Q/\sigma$. The holomorphic **16**-bundle $V_{16} \rightarrow \hat{Q}$ is the continuum limit of the cellular cosheaf on the 1|10|16 partition.

Theorem 5.9 (Observer fixed-point theorem). *Let $\bar{\partial}_{V_{16}}$ be the Dolbeault operator on (Q, V_{16}) and σ its antiholomorphic lift. The equivariant Lefschetz index*

$$\text{ind}_\sigma(\bar{\partial}_{V_{16}}) = \text{Tr}_{H^1(Q, V_{16})}(\sigma) = -3 \quad (5.5)$$

has $|\text{ind}_\sigma| = 3$ anti-invariant fixed-point contributions, each an observer slot. Moreover,

$$\dim H^1(\hat{Q}, V_{16})^{\sigma=-1} = 3, \quad (5.6)$$

and the map $\iota \mapsto |\omega_\iota\rangle$ is a bijection between $\mathcal{I}$ and a basis of this space.

Proof. Cellular count on $\mathcal{T}_7$ (Appendix A.2); Atiyah–Bott equivariant index [7]; passage to $H^1(\hat{Q}, V_{16})$ via the $\mathbb{Z}_2$ quotient. $\square$

Corollary 5.10 (Observer from fixed points). *An observer is a fixed-point label of σ on $H^1(Q, V_{16})$: not a point of Q fixed by σ (there are none in the **16**-sector), but an anti-invariant cohomology class whose Lefschetz contribution is -1 . The fixed-point theorem is the clincher: it converts global mirror data into exactly three individual observer slots.*

5.5 Lifetime chart: index to time

An observer index ι is static data. A *lifetime* is a trajectory inside slot ι , charted by a time parameter.

Definition 5.11 (Lifetime map). Fix $\iota \in \mathcal{I}$. A *lifetime chart* is a pair

$$(\tau_\iota, g_\iota), \quad g_\iota : \mathbb{R} \longrightarrow \mathcal{S}_\iota, \quad (5.7)$$

where τ_ι is the observer's time parameter and $\mathcal{S}_\iota$ is the soldered state bundle over the ι -sector: sections of $V_{16}|_{\hat{Q}}$ restricted to the ι -diagonal cell, with values in the order parameter Φ (Section 6.2). The map g_ι is *admitted* by the capacity flow on $\mathcal{G}_{\text{Albert}}$ when

$$\frac{d}{d\tau_\iota} g_\iota(\tau_\iota) = -K^{-1} \nabla V(g_\iota(\tau_\iota))|_\iota + \partial_{\tau_\iota} \sigma|_{H_2(\mathbb{H})}, \quad (5.8)$$

with K the H -invariant coset metric and the second term the soldering clock in the quaternionic block.

Definition 5.12 (Lifetime index). The *lifetime index* is

$$\Lambda(\iota, \tau) := (\iota, \tau, \mathcal{C}_\iota(\tau)), \quad \mathcal{C}_\iota(\tau) := \omega_\iota^*(g_\iota(\tau)), \quad (5.9)$$

recording which observer (ι), which time (τ), and which capacity cell the trajectory occupies. Two lifetimes are the same iff their indices Λ agree.

Theorem 5.13 (Index–lifetime mapping). *For each $\iota \in \mathcal{I}$ there exists a maximal lifetime interval $I_\iota \subset \mathbb{R}$ and a chart $g_\iota : I_\iota \rightarrow \mathcal{S}_\iota$ satisfying (5.8) with initial condition $g_\iota(\tau_{\text{birth}}) = \Phi_\star|_\iota$. The correspondence*

$$\iota \longmapsto (\tau_{\text{birth}}, \tau_{\text{death}}, g_\iota) \quad (5.10)$$

is functorial under mirror descent $E_8 \times E_8 \rightarrow E_6$: capacity collapse above determines the admissible time range I_ι below.

Proof sketch. Existence of g_ι is standard gradient flow of V on the ι -restricted coset. Functoriality follows because $\text{Bal}_\mathcal{C}$ at $n = 8$ fixes $\omega^\star$ and hence the initial datum $\Phi_\star|_\iota$; soldering (Definition 6.3) supplies the clock direction $\partial_{\tau_\iota} \subset H_2(\mathbb{H})$. $\square$

Conjecture 5.14 (Uniqueness of lifetime chart). *Given ι and $\Phi_\star|_\iota$, the lifetime chart (τ_ι, g_ι) is unique up to reparametrization of τ_ι among all F_4 -equivariant flows compatible with mirror functoriality and the above–below correspondence (5.1).*

Remark 5.15 (Arbitrary input and determinism). The void and mirror fix the observer *slot space* $\mathcal{I}$ and the dynamics (5.8). Choosing $\iota \in \mathcal{I}$ is the minimal arbitrary input identifying *who* observes; choosing $\tau \in I_\iota$ identifies *when* within that lifetime. No further postulate of a conscious subject is introduced: an observer *is* the collapsed anti-invariant fixed-point class with its admitted time chart.

6 The Physical Universe

Mathematical data of $\mathfrak{U}^\star$ becomes physical through mirror quotient observability (Section 5), observer collapse at $M^7(\mathcal{V})$, and soldering on the Jordan–octonionic block.

6.1 Physical sector

Physical configurations are observer-collapsed states on $\hat{Q}$ (Definitions 5.8, 5.6, 6.1).

Definition 6.1 (Physical configurations). A configuration is *observable* if it lies in the σ -anti-invariant sector of $\hat{Q} = Q/\sigma$. Fermions are cohomology classes in $H^1(\hat{Q}, V_{16})^{\sigma=-1}$; each observer $|\omega_i\rangle$ is one such class (Theorem 5.9).

Theorem 6.2 (Three families). $\dim H^1(\hat{Q}, V_{16})^{\sigma=-1} = 3$. Each class is one chiral **16** of $\text{Spin}(10)$ —one observer slot $\iota \in \mathcal{I}$; mirror partners are σ -invariant and non-observable. Hence $n_F = 3$.

Proof. Theorem 5.9; cellular details in Appendix A.2. $\square$

6.2 Spacetime from the climb

Definition 6.3 (Soldering). Embed $\mathbb{H} \subset \mathbb{O}$ from the mirror-fixed associative subalgebra at the Jordan exit. The block $H_2(\mathbb{H}) \subset H_2(\mathbb{O}) \cong \mathbb{R}^{1,9}$ carries Lorentzian signature via $\det$. The soldering map σ identifies four clock components $\Phi^0, \dots, \Phi^3$ with this block; $d = 4$.

Proposition 6.4 (Coset metric). $K|_{\mathfrak{m}}$ is positive-definite and H -invariant. Lorentzian signature is carried by σ , not by Wick rotation of K .

Theorem 6.5 (Semiclassical reflection positivity). On *EIII*, with radiative mass $\mu^2 > 0$ on 28 coset scalars (Coleman–Weinberg), the Euclidean σ -model (4.4) satisfies reflection positivity along Φ^0 in the semiclassical regime.

Conjecture 6.6 (Nonperturbative reflection positivity). Tree-level Goldstones and the nonperturbative gauge measure satisfy full reflection positivity, yielding OS reconstruction [9].

6.3 Gauge, anomalies, gravity

Theorem 6.7 (Anomaly cancellation). For one embedded **27**: cubic $16 - 80 + 64 = 0$; mixed coefficient -1 cancelled by Green–Schwarz [8]. With $n_F = 3$ replicas of the full **27** arithmetic, anomalies cancel family-wise.

Theorem 6.8 ($b_0 = 12$). With $C_A(\text{Spin}(10)) = 8$, $T(\mathbf{16}) = 2$, $n_F = 3$, $n_S = 28$: $b_0 = (88 - 24 - 28)/3 = 12$.

Theorem 6.9 (Induced Planck mass). Sakharov induction from 28 massive coset scalars gives $M_{\text{Pl}}^2 \propto 28\mu^2/(6 \cdot 16\pi^2) \log(\Lambda^2/\mu^2)$.

Theorem 6.10 (Gravitational fixed point). One-loop truncation gives $\kappa_\star = 96\pi^2/5$ with $\partial\beta_\kappa/\partial\kappa|_{\kappa_\star} = -2$.

Conjecture 6.11 (GUT breaking). Gradient flow of V on coset directions triggers $E_6 \rightarrow \text{Spin}(10) \times U(1) \rightarrow \dots \rightarrow SU(3)_c \times U(1)_{\text{em}}$ with Higgs content **45**_H, **126**_H, **10**_H.

7 Conclusions

SJET builds the universe from the mirror:

$$\mathcal{V} \xrightarrow{\text{M}} \dots \xrightarrow{\text{M}} \mathfrak{U}^\star, \quad \mathfrak{U}^\star = C^\infty(\mathcal{V}). \quad (7.1)$$

Observers arise at the heterotic rung as collapsed fixed-point states:

$$\iota \mapsto |\omega_\iota\rangle \in H^1(\hat{Q}, V_{16})^{\sigma=-1}, \quad g_\iota : \mathbb{R} \rightarrow \mathcal{S}_\iota. \quad (7.2)$$

Only two axioms are primitive. Duality, observer multiplicity, and the index–lifetime map are **derived** from $E_8 \times E_8$ collapse and the Lefschetz fixed-point theorem.

7.1 Epistemic summary

Status	Content
Axioms	void, mirror
Theorems	rung table, E_8 , $E_8 \times E_8$, E_6 , partition, vacuum, duality, observer fixed points, three families, index–lifetime map, anomalies, $b_0 = 12$, $\kappa_\star$, semiclassical RP
Definitions	above–below, observer, lifetime chart, physical sector
Conjectures	lifetime uniqueness, full RP, GUT breaking, CC sequestering

7.2 Open problems

- O1.** Prove Conjecture 5.14 (functorial lifetime chart).
- O2.** Functorial uniqueness of soldering (Definition 6.3).
- O3.** Prove Conjecture 6.11 from V .
- O4.** Prove Conjecture 6.6.
- O5.** Graviton/ R^2 persistence of $\kappa_\star$.
- O6.** Cosmological constant from σ -invariant sequestering at $\kappa_\star$.

7.3 Archive

The prior modular manuscript (39 pages, verification ledger, axiomatic elimination program) is preserved in `old/sjet-stack/`.

A Albert algebra and mirror quotient

A.1 Albert model

Elements of $J_3(\mathbb{O})$ are 3×3 Hermitian octonionic matrices

$$X = \begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \quad \xi_i \in \mathbb{R}, \ x_i \in \mathbb{O}. \quad (\text{A.1})$$

The sharp product $X^\#$ and cubic norm $N(X) = \det X$ define the E_6 structure group [5, 6].

A.2 Mirror quotient cohomology

On the stabilized tower $\mathcal{T}_7$ at $M^7(\mathcal{V})$, let σ act on the rank-3 diagonal $\mathfrak{d}$ by $\sigma(e_i) = -e_i$ with triality rotation on the three **16**-cells (C_1, C_2, C_3) .

Lemma A.1 (Cellular observer count). *With $\mathcal{F}_{16}$ the **16**-cellular cosheaf on $\mathcal{T}_7$,*

$$\dim H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=+1} = 0, \quad \dim H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=-1} = 3. \quad (\text{A.2})$$

Each $\sigma = -1$ class is one observer index $\iota \in \{1, 2, 3\}$.

Proof. Cellular 1-cochains are triples $(\phi_1, \phi_2, \phi_3) \in (\mathbb{C}^{16})^3$ with cocycle constraint $\phi_1 + \phi_2 + \phi_3 = 0$ at the **10**-hub. Coboundaries are diagonals (ψ, ψ, ψ) . The quotient is two copies of $\mathbb{C}^{16}$ as a σ -module; triality with sign leaves three anti-invariant classes, one per diagonal direction e_i . $\square$

Proposition A.2 (Stone lift). *The profinite lift to Q identifies $H^1(\mathcal{T}_7, \mathcal{F}_{16})^{\sigma=-1} \cong H^1(Q, V_{16})^{\sigma=-1}$.*

Theorem A.3 (Mirror-quotient Lefschetz index). *With $\bar{\partial}_{V_{16}}$ on (Q, V_{16}) and antiholomorphic σ ,*

$$\text{ind}_\sigma(\bar{\partial}_{V_{16}}) = \text{Tr}_{H^1(Q, V_{16})}(\sigma) = -3, \quad \dim H^1(\hat{Q}, V_{16})^{\sigma=-1} = 3. \quad (\text{A.3})$$

This is Theorem 5.9; each unit contribution is one observer fixed-point slot.

Proof. Lemma A.1, Proposition A.2, and the Atiyah–Bott holomorphic Lefschetz formula [7]. For a $\mathbb{Z}_2$ quotient by an antiholomorphic involution, $H^1(\hat{Q}, V_{16}) = H^1(Q, V_{16})^{\sigma=-1}$. $\square$

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